Mücahid Enes Deniz

Software Engineer

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Summary

Software engineer with hands-on experience developing and deploying industrial software using C++ and Python. Built computer vision and automation systems used across three factories, with a strong interest in AI and machine learning.

Education

- **Bahçeşehir University** — Istanbul, TR
  B.Sc., Computer Engineering (Full Scholarship) | GPA: 3.07/4.00 — *Sep 2020 - Jun 2025*
- **Bornova Anatolian High School** — Izmir, TR
  Science & Math | GPA: 93.3/100 — *Sep 2016 - Jun 2020*

Work Experience

- **Kastamonu Entegre (Hayat Holding)** — Istanbul, TR
  Software Engineer (Full Time) — *Jul 2025 - Present*
  - **Patented Fiber Analysis**: Developed and deployed C++/Python vision software analyzing MDF fibers across three factories, reducing CAPEX by 88% versus the commercial alternative. Connected Qt and Python through Redis IPC; automated PLC driven lens adjustments and device recovery.
  - **Parquet Defect Inspection**: Delivered AI inspection for parquet undersides outside the existing system's coverage. Built a five-stage acquisition, stitching, and inference pipeline with bounded queues and defect alerts; accepted for production use.
  - **Panel Geometry Inspection**: Implemented submillimeter diagonal and squareness measurement using dual Basler cameras, OpenCV, and PLC derived calibration, with Redis/SQL records for quality tracking.
  - **Truck Load Measurement**: Fused four SICK LiDAR streams with motion compensation for truck cargo volume estimation; integrated Flask/PostgreSQL dashboards and weighbridge software to automate manual measurement.
  - Management projected **EUR 11.57M** combined financial impact upon planned rollout of these four projects, including capital expenditure savings and operational benefits.

  TalenTree Long Term Intern — *Jul 2024 - Jun 2025*
  - Supported industrial AI software development and deployment; served as Scrum Master on a Sustainable Supplier Selection project.
- **Harpeon**
  Mobile Application Development Intern — *Jun 2023 - Aug 2023*
  - Built an API integrated weather app and an office reservation UI prototype using React Native and Expo.

Projects

- **Price Tracking Platform (Python, FastAPI, PostgreSQL, React)**: Built a self-hosted platform with background workers and Telegram alerts, Docker Compose deployment, automated tests, and GitHub Actions CI.
- **Blast Style Puzzle Game (Unity, C#)**: Implemented BFS-based matching, rocket power-ups, obstacle-aware gravity, and JSON defined levels.
- **PAC-MAN Clone (Unity, C#)**: Built buffered grid movement and four-state ghost AI with BFS pathfinding and line-of-sight chase detection.

Skills

Languages: C++, Python, C#, JavaScript, SQL
Systems & Backend: Qt, FastAPI, Flask, Redis, PostgreSQL, multithreading, IPC
Computer Vision: OpenCV, YOLO, ONNX Runtime, OpenVINO
Testing & Deployment: Git, Docker Compose, GitHub Actions, CMake, CTest
Industrial Integration: Basler Pylon, SICK LiDAR, Siemens S7 / Snap7, OPC UA
Game Development: Unity, Godot

Languages & Interests

Languages: Turkish (Native), English (Advanced)
Interests: Gaming and esports, chess, salsa, running, taekwondo, hiking.

Last Updated: September 2026